

Chasing Zero: Why Faster Speeds Don't Always Save Time

The Law of Diminishing Returns in Travel Time

Mathematics Exploration

2026-06-16

Setup

The formula for average speed is given by:

$$s = \frac{d}{t}$$

We can rearrange this formula to find the time (t) taken to travel a distance (d) at a constant speed (s):

$$t = \frac{d}{s}$$

Suppose we want to find the time saved when travelling the same distance d at two different average speeds, s_1 and s_2 . Assuming s_2 is the faster speed ($s_2 > s_1$), the time difference (T) is expressed as:

$$\begin{aligned} T = t_1 - t_2 &= \frac{d}{s_1} - \frac{d}{s_2} \\ &= d \left(\frac{1}{s_1} - \frac{1}{s_2} \right) \end{aligned}$$

Examples and Analysis

To illustrate the mathematical relationship between speed and time, let $d = 40$ miles (roughly the distance between Manchester and Leeds in the UK). By observing different speed increments, we can see a clear law of diminishing returns: as initial speed increases, equal increments of additional speed yield smaller and smaller time savings.

Case 1: Driving Speeds

First, consider a low-speed environment. Increasing a constant speed from 40 mph to 50 mph results in a substantial gain:

$$T = 40 \left(\frac{1}{40} - \frac{1}{50} \right) = 0.2 \text{ hours} = 12 \text{ minutes}$$

However, as the baseline speed increases, the time savings begin to compress. Using the same distance of 40 miles:

- Increasing from 50 mph to 60 mph saves approximately **8 minutes**.
- Increasing from 60 mph to 70 mph saves approximately **5.71 minutes**.
- Increasing from 70 mph to 80 mph saves approximately **4.29 minutes**.

Notice that pushing from 70 to 80 mph saves less than half the time that was saved by pushing from 40 to 50 mph, despite it being the exact same 10 mph increment.

Interestingly, to achieve another significant chunk of time savings at higher speeds, the velocity jump must be much larger. For example, pushing from 80 mph to an extreme 100 mph (a 20 mph jump) saves exactly **6 minutes**.

Case 2: Jet Travel

This phenomenon becomes extreme when dealing with aviation speeds. Suppose we travel this same 40 miles by air. One jet cruises at 500 mph and a much faster jet flies at 1000 mph.

Even though the second jet has doubled its speed—adding an astronomical 500 mph—the baseline speed is already so high that the journey is completed almost instantly by both. Thus, the actual time saved is nominal:

- Increasing from 500 mph to 1000 mph saves exactly **2.4 minutes**.

Case 3: Jet Travel vs Speed of Light

To push this concept to its absolute logical extreme, imagine comparing a commercial jet flying at 500 mph to a hypothetical vehicle travelling at the speed of light ($c \approx 670,616,629$ mph).

At 500 mph, a 40-mile trip takes a mere 4.8 minutes (288 seconds). At the speed of light, the trip is instantaneous, taking roughly 0.000215 seconds. Despite an unimaginable increase in speed of over 670 million miles per hour, the absolute maximum amount of time you can possibly save over this distance is capped by the jet's already brief flight time:

- Increasing from 500 mph to the speed of light saves approximately **4.8 minutes**.

This demonstrates that once travel time approaches zero, further speed increases become practically irrelevant for local commuting. Mathematically, this behavior occurs because the fast time component approaches a limit of zero:

$$T \approx \frac{d}{s_1} \quad \text{as} \quad \frac{d}{s_2} \rightarrow 0$$

Conclusion

Because time is inversely proportional to speed, the relationship is non-linear. Travel time drops sharply at low speeds, but flattens out at high speeds. Therefore, over short distances, massive increases in high speeds produce negligible practical benefits.